

PnL Class Cheat Sheet

 `PnL(*, name, source, groups=None, role=None, consideration=None, obligation=None, scale=None, result_name='result', label=None, label_map=None)`

Two construction forms: the ledger `groups=[Group(...), ...]`, and the sugar `role=/consideration=/obligation=` for exactly one Group. A  `Group(label, role, consideration, obligation)` holds  `Leg(label, func, bs, is2d)` cash flows.

A PnL is a position: one stochastic source plus an ordered ledger of signed Groups. Each group books a consideration and an obligation under a role (sell → +consideration, -obligation; buy → the contra), so the EX column adds straight down the sheet. All derived rows are per-atom partial sums, never arithmetic on marginals, so covariance is carried for free. Build with `build('pnl ...')` / `build('xpnl ...')`, or  `Aggregate.make_pnl`. `pnl` is the *consolidated* single-group net view, a flat Consideration / Obligation / Margin card; `xpnl` is the exploded sibling, same syntax, a multi-group walk with one group per step, so `summary_df` carries one (Step, View) block per step plus a closing 'All', and `stats_df` is three-level (Step, View, Line). The following tables show all  methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

`name` (identity), `label` (display), `source` (the stochastic generator: the *simplest sufficient* object, so a `GridDistribution`, an `Aggregate`, a bare (values, probs) pair, or a `BivariateDistribution` joint), `engine` (the full `Aggregate` / `Portfolio` behind it, for drill-down), `groups` (the declared Group specs, in ledger order), `role` (the single group's role; `None` on a multi-group ledger), `result_name`, `economics` (the resolved cession dict from `deposit` / `rol` / `rate` / `cede`), `program`, `pprogram`, `construction_description`, `construction_explanation` (how this P&L was built)

Two P&Ls over the same source concatenate with + (their ledgers chain).

2. Update

None. A PnL is a value object: it is assembled once from an already-updated source and holds signed per-atom rows plus their exact moments, with no retained engine dependency. Re-declare (or rebuild the engine) to change it.

scale is the one committed setting: the unitizer for `summary_df` and `scaled_stats_df` (default: the expected signed grand total consideration).

3. Moments

`est_m` ($E[\text{result}]$, the signed grand net), `est_sd`, `est_cv` (meaningless near break-even, where the mean passes through zero; reported for symmetry), `est_skew`, `E_consideration` ($E[\text{total consideration}]$, signed across groups)

Per-row moments (every leg, group total, group result, running net, and the grand rows) are the EX / SD / CV / Skew columns of `stats_df`.

4. Statistical functions

 `cdf` ($\Pr(\text{result} \leq x)$),  `sf`,  `q` (quantile = VaR of the result),  `tvar`, `prob_eq_0` ($\Pr(\text{exact break-even})$), `result`, `gd` (the grand result as a `GridDistribution`; `gd` is an alias)

All of these read the grand result, the signed bottom line of the ledger. Adverse outcomes sit in the same tail for either role.

5. Validation

`validation_df` (Est-vs-EX audit for every `bs > 0` leg) *EX is the exact signed mean straight off the atoms; Est the mean of the rebucketed grid. The linear scheme matches means, so this reads very close, but it is visible. Legs with `bs = 0` are exact and absent. No reaching through to the source's own validation.*

6. Output dataframes

`summary_df` (the headline card: fixed rows that never vary with the ledger), `stats_df` (the full ledger × metrics sheet, in currency: the alignment / footing exhibit), `scaled_stats_df` (the same, unitized by scale), `density_df` (an ordered {row label: `GridDistribution`} — *not* one stapled frame; each row keeps its own exact grid), `info`, `figure`

7. Reinsurance

`economics` (the resolved `pc_occ` / `pc_agg` / `c_occ` / `c_agg` / `gross` / `ceded` dict)

A cession is an ordinary group booked buy: `consideration = ceded premium`, `obligation = recovery + commission`. Reinstatements and variable rating (swing / slide / `pc` / corridor) are declared on the layer and consolidate into the same ledger; the resolved contract objects live on `engine.reinstatement_terms` / `engine.variable_terms`.

8. Visualization

 `plot` (grand result density and CDF; iterates `density_df`), `figure`

9. Risk and pricing

 `evaluate` (the Cherny–Madan breakeven acceptability panel)

A P&L is evaluated, not priced. You price the obligation and ask which distortion drives the risk-adjusted net to zero: the breakeven stress the position survives. One row per family (`ph`, `wang`, `dual`, `tvar`), with `param`, `error`, `gini_p`, `area`. The breakeven `gini_p` is the single family-agnostic acceptability index.

Needs one obligation leg over a regular lattice source.

10. Approximations

None. The ledger is exact on the source atoms; the only discretization is each leg's own rebucketing onto its grid, audited by `validation_df`.

11. Meta

 `help`,  `format_program`, `pprogram_html`, `note`, `tags`, `doc`, `hints` (the DecL trailer), `label_map`, `labels`, `renamer`, `use_labels`